

Software Engineering Lab - Lab 1

Requirements Engineering & UML Use-Case Modelling

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Problem Statement #48 - Incident Escalation & On-Call Rotation Engine

System overview: A DevOps on-call system for rotations, incident alert escalation and post-mortem tracking.

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest incident alerts, identify the active on-call engineer, and escalate to the secondary engineer if the alert is not acknowledged within 5 minutes.	High	Pass: escalation starts after the 5-minute timeout. Fail: the alert remains unacknowledged without escalation.	This is the main purpose of the system.
FR-002	Functional	The system shall allow the Incident Commander to create and update multi-tier escalation rules, notification channels, and timeout values.	High	Pass: a saved rule is used for the next matching alert. Fail: the old rule is still used.	Different teams need different escalation policies.
FR-003	Functional	The system shall allow an on-call engineer to acknowledge, reassign, or resolve an incident from the dashboard, SMS, or phone response.	High	Pass: a valid action updates the incident status within 10 seconds. Fail: the status does not change.	Engineers may need to respond while away from a laptop.
FR-004	Functional	The system shall maintain recurring on-call schedules with shift changes, manual overrides, and holiday substitutions.	Medium	Pass: the correct engineer is returned for normal and override time periods. Fail: the wrong engineer is selected.	The active responder must be known at any time.
FR-005	Functional	The system shall create a post-mortem record when an incident is resolved, including the event timeline, root-cause field, and action items.	Medium	Pass: a post-mortem is created and linked to the resolved incident. Fail: no linked record is created.	Post-mortems help the team review incidents later.
NFR-001	Nonfunctional	Alert dispatch through webhook, SMS, and email must start within 3 seconds of alert ingestion.	High	Pass: measured dispatch start time is 3 seconds or less under the simulated peak load. Fail: it is above 3 seconds.	Fast notification reduces response delay.
NFR-002	Nonfunctional	The system shall target 99.9% availability and protect incident/contact data in transit and at rest.	High	Pass: availability meets 99.9% and no unencrypted data path is found. Fail: either condition is not met.	The alerting system must remain available and protect contact data.

2. UML Use-Case Diagram

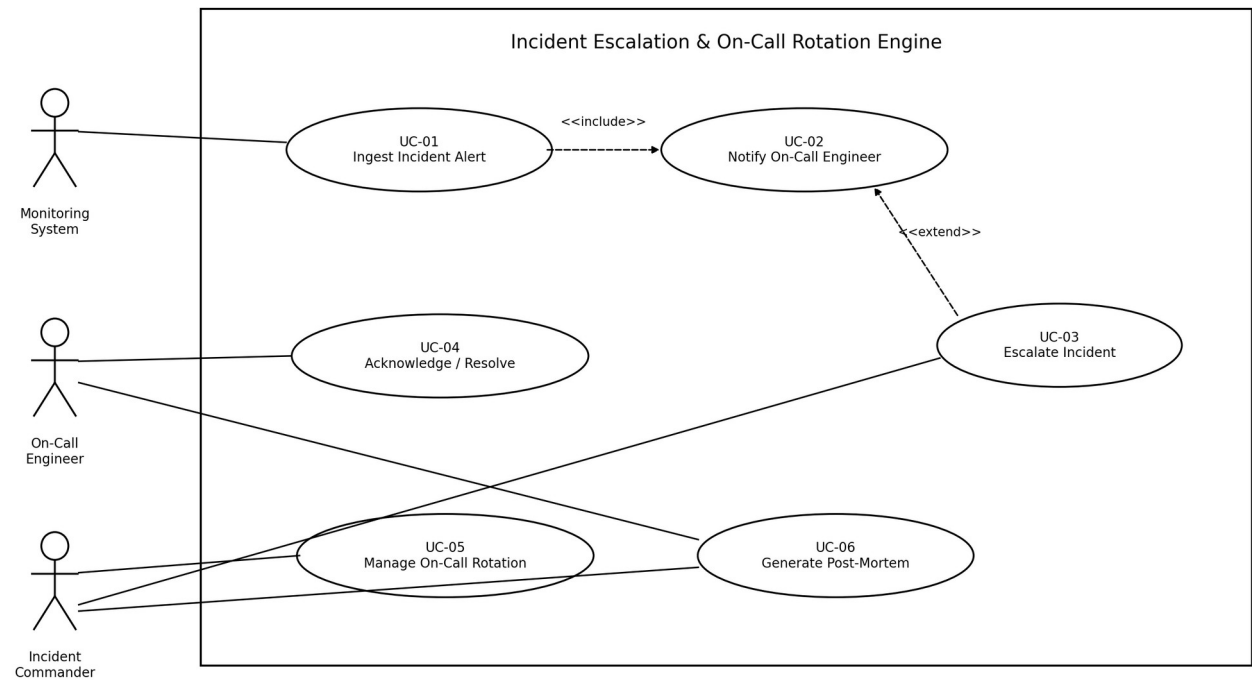


Figure 1. UML use-case diagram with three actors, six use cases, <<include>> and <<extend>>.

3. Use-Case Flow - UC-01: Ingest Incident Alert

Preconditions: an on-call schedule and an escalation policy are configured; the monitoring webhook is active.

Postconditions: the incident is recorded and its status is either acknowledged or escalated.

Step	Main Success Scenario
1	Monitoring system sends an incident alert to the webhook endpoint.
2	System validates the alert and creates an incident record.
3	System checks the on-call schedule and finds the current Tier-1 engineer.
4	System reads the escalation policy and selected notification channels.
5	System sends the alert to the Tier-1 engineer.
6	The engineer acknowledges the incident from the dashboard, SMS, or phone.
7	System records the acknowledgement and stops the escalation timer.
8	The incident remains available for investigation until it is resolved.

Alternate Flow - No acknowledgement:

1) The 5-minute timeout expires without acknowledgement. 2) System marks the incident as escalated and selects the Tier-2 engineer. 3) System notifies Tier-2 and informs the Incident Commander. 4) When Tier-2 acknowledges, the timer is stopped and the incident continues normally.